



Assignment_02

MAT216_10_13

Spring 2024

Remarks: Release date: 23.2.2024

Submission date: Sec_10 (3.3.2024, Sunday, Class time)

Sec_13 (4.3.2024, Monday, Class time)

Total marks: 40 (will be converted to 20)

1. [10]

Suppose $\text{rref}(R_0) = A$ where $R_0 = \begin{bmatrix} 1 & 3 & 5 & 0 & 7 \\ 2 & 6 & 10 & 1 & 16 \\ 3 & 9 & 15 & 1 & 23 \end{bmatrix}$. Show that –

- a) The row space has dimension 2, matching the rank
- b) The column space of R_0 has also dimension $r = 2$
- c) The null space of R_0 has dimension 3
- d) The null space of R_0^T , which can also be called the left null space of R_0 ; has dimension

2. [10]

Let $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ and consider the following subset V of the 2-dimensional vector space R^2 , Namely $V = \{x \in R^2 | Ax = 5x\}$

- a) Prove that the subset V is a subspace of R^2
- b) Find a basis for V and determine the dimension of V

3. Determine the rank of the matrix A:

[10]

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

4. Let W be the set of all points from $\mathbb{R}^3$ of the form $(1, x_2, x_3)$. Is this a subspace of $\mathbb{R}^3$? [5]

5. **Vector Space:** A set V equipped with two binary operations addition and scalar multiplication is called a vector space over the field F , if V satisfies the following 10 axioms, [5]

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| (i) $u + v \in V$ for all $u, v \in V$ | (vi) $ku \in V$ for all $u \in V$ and $k \in F$ |
| (ii) $u + v = v + u$ | (vii) $a(u + v) = au + av$ for all $a \in F$ and $u, v \in V$ |
| (iii) $u + (v + w) = (u + v) + w$ | (viii) $(a + b)u = au + bu$ for all $a, b \in F$ and $u \in V$. |
| (iv) There exists a $0 \in V$ s. t. $u + 0 = u$ for all $u \in V$ | (ix) $(ab)u = a(bu)$ for all $a, b \in F$ and $u \in V$ |
| (v) There exists a $-u \in V$ for all $u \in V$, such that $u + (-u) = 0$. | (x) $1u = u$, where $1 \in F$ and for all $u \in V$. |

(a) Suppose $u = (u_1, u_2)$ the multiplication of cu is defined to produce $(cu_1, 0)$ instead of (cu_1, cu_2) . With usual addition in $\mathbb{R}^2$, which of the eight conditions are not satisfied.?